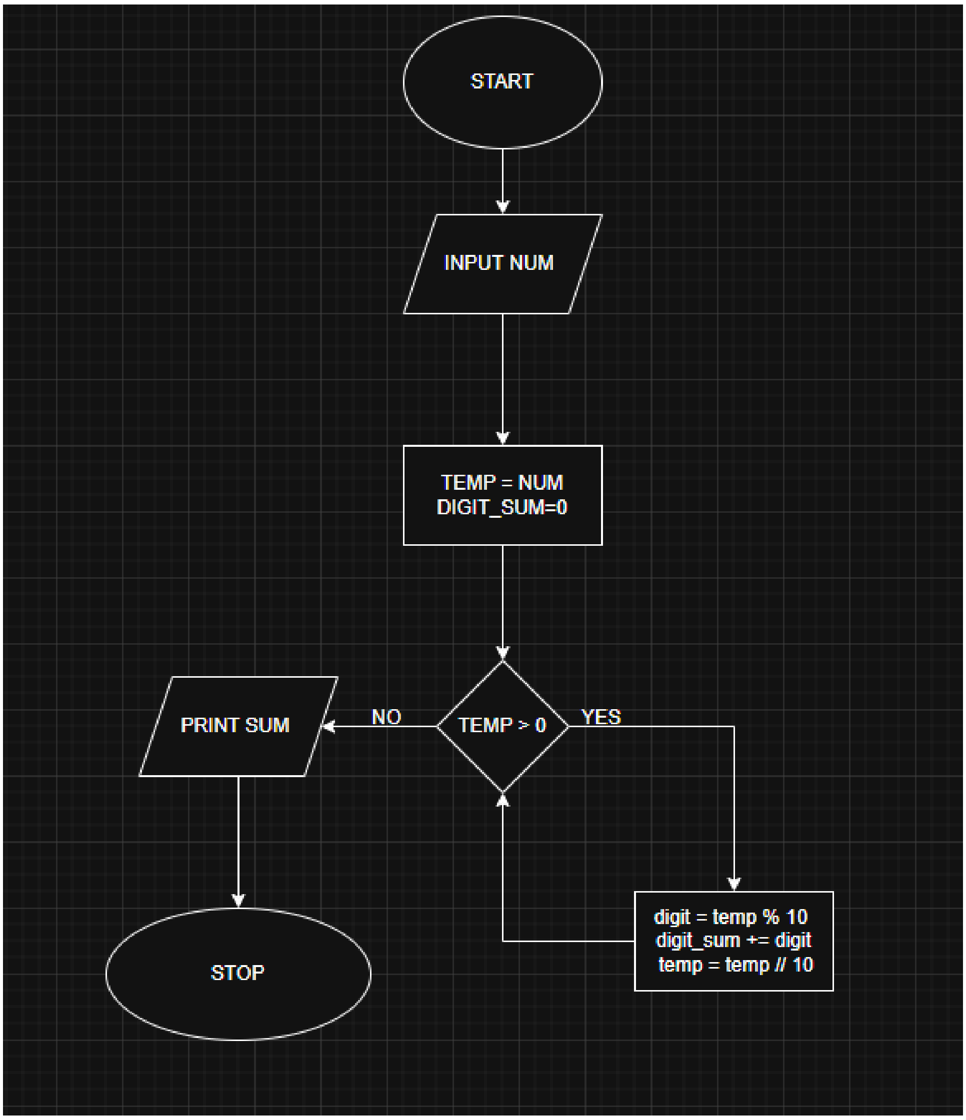
11.1.1 SUM OF DIGIT OF NUMBER

ALGORITHM

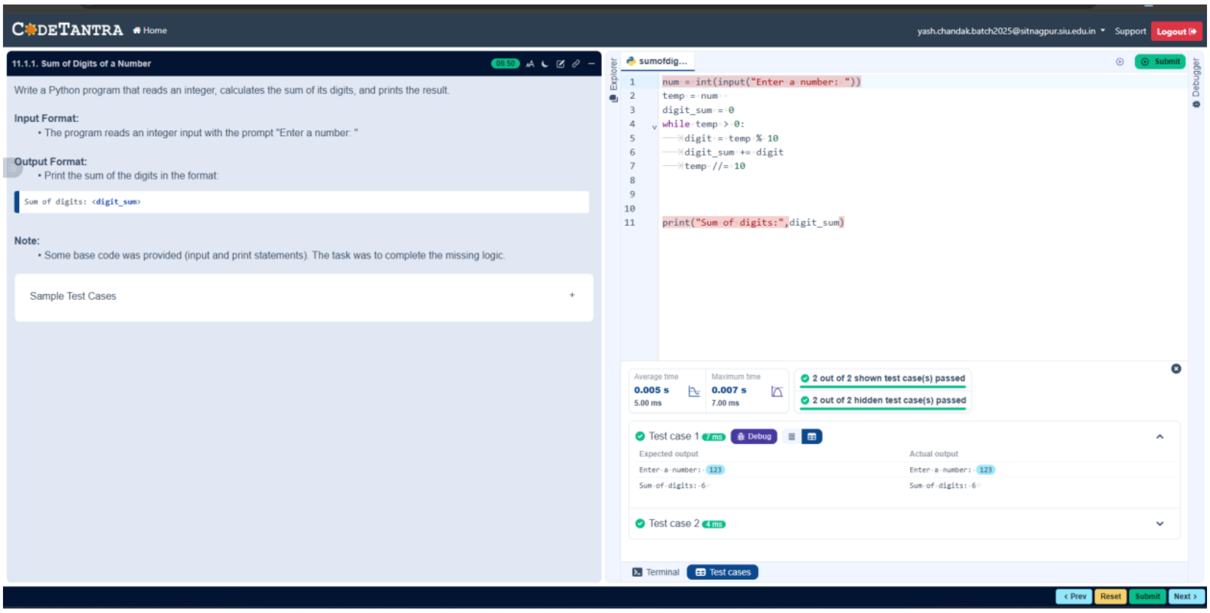
1. Start
2. Input num
3. Set temp = num, digit_sum = 0
4. While temp > 0:
   digit = temp % 10
   digit_sum = digit_sum + digit
   temp = temp // 10

5. Print digit_sum
6. End

FLOWCHART



PROGRAM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yash.chandak.batch2025@sitnagpur.siu.edu.in
Support
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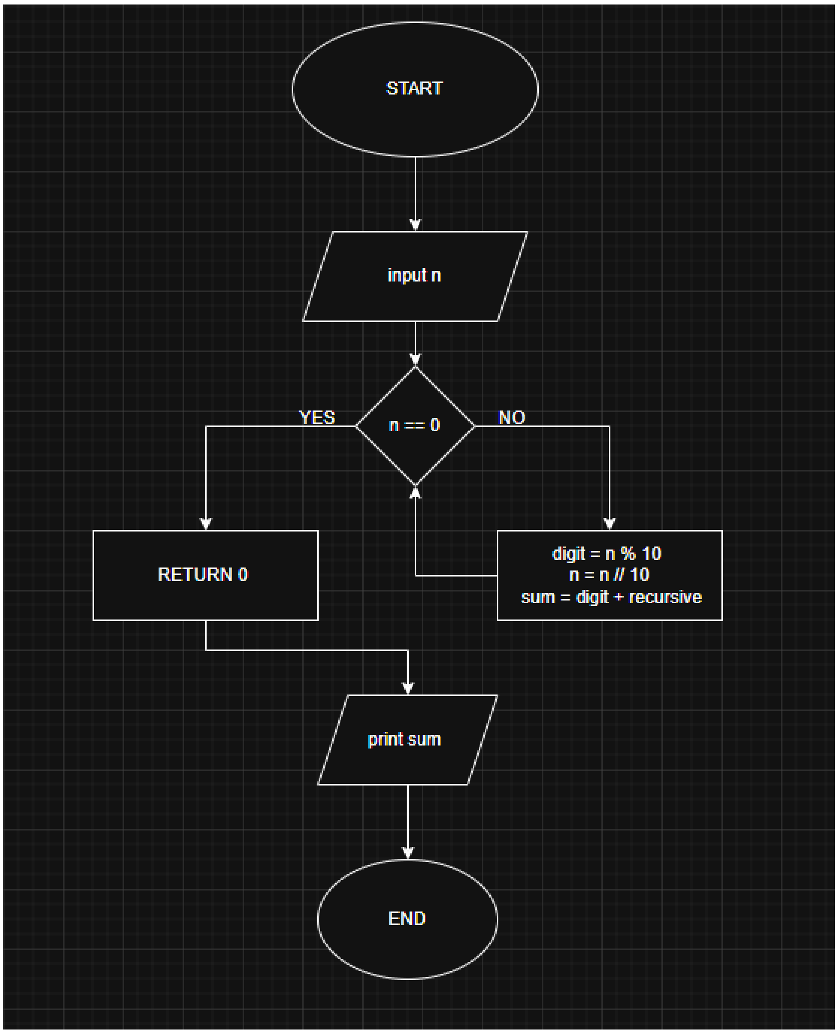
11.1.1. Sum of Digits of a Number
06:50
Write a Python program that reads an integer, calculates the sum of its digits, and prints the result.
Input Format:
• The program reads an integer input with the prompt "Enter a number: "
Output Format:
• Print the sum of the digits in the format:
Sum of digits: <digit_sum>
Note:
• Some base code was provided (input and print statements). The task was to complete the missing logic.
Sample Test Cases
Explorer
sumofdig...
Submit
Debugger
num = int(input("Enter a number: "))
temp = num
digit_sum = 0
while temp > 0:
digit = temp % 10
digit_sum += digit
temp //= 10
print("Sum of digits:",digit_sum)
Average time
0.005 s
5.00 ms
Maximum time
0.007 s
7.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
7 ms
Debug
Expected output
Enter a number: 123
Sum of digits: 6
Actual output
Enter a number: 123
Sum of digits: 6
Test case 2
4 ms
Terminal
Test cases
Prev
Reset
Submit
Next

11.1.2 SUM OF DIGIT USING RECURSION

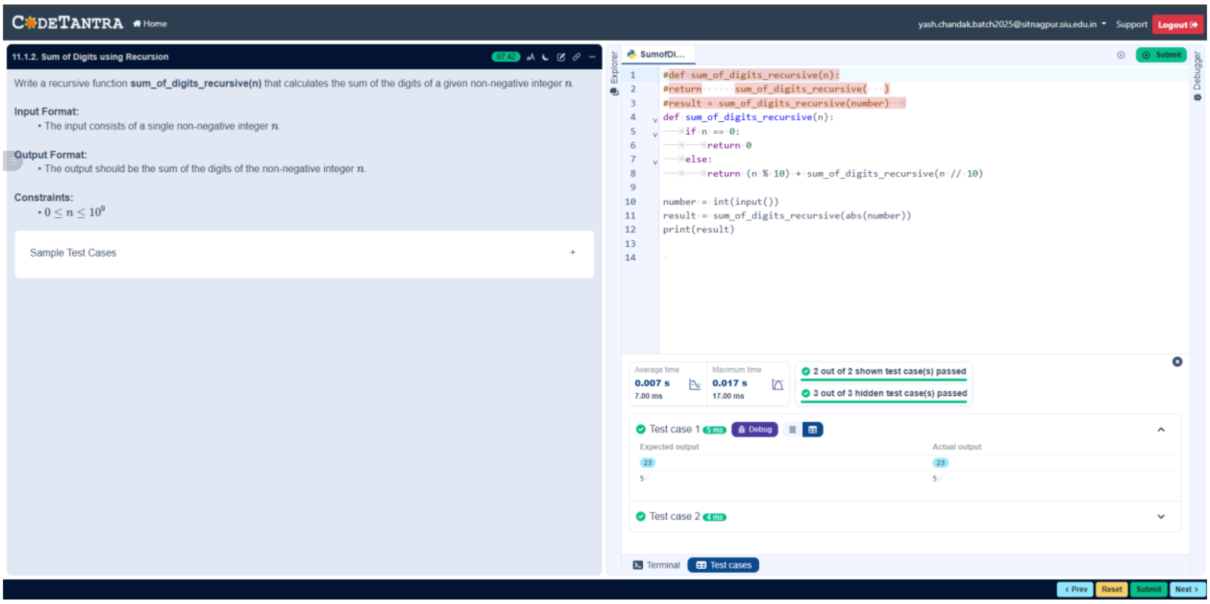
ALGORITHM

Step 1: Start
Step 2: Input integer `n`
Step 3: Check if `n == 0` If yes → return `0`
Step 4: Otherwise
Compute `(n % 10)` → last digit
Compute `sum_of_digits_recursive(n // 10)` → recursive call
Step 5: Return `(n % 10) + sum_of_digits_recursive(n // 10)`
Step 6: Print result
Step 7: End

FLOWCHART



PROGRAM



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11.1.2. Sum of Digits using Recursion
07:42
Write a recursive function sum_of_digits_recursive(n) that calculates the sum of the digits of a given non-negative integer n.
Input Format:
• The input consists of a single non-negative integer n.
Output Format:
• The output should be the sum of the digits of the non-negative integer n.
Constraints:
• 0 ≤ n ≤ 10^9
Sample Test Cases
Explorer
SumofDi...
Submit
Debugger
#def sum_of_digits_recursive(n):
#return sum_of_digits_recursive( )
#result = sum_of_digits_recursive(number)
def sum_of_digits_recursive(n):
if n == 0:
return 0
else:
return (n % 10) + sum_of_digits_recursive(n // 10)
number = int(input())
result = sum_of_digits_recursive(abs(number))
print(result)
Average time
0.007 s
7.00 ms
Maximum time
0.017 s
17.00 ms
2 out of 2 shown test case(s) passed
3 out of 3 hidden test case(s) passed
Test case 1
5 ms
Debug
Expected output
23
5
Actual output
23
5
Test case 2
4 ms
Terminal
Test cases
‹ Prev
Reset
Submit
Next ›